TechTones Brand Sentiment Analysis

Classifying Public Perception of Apple vs. Google Using NLP

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Project Goal and Key Outcomes

Business Objective:

- Develop a high-accuracy Natural Language Processing (NLP) classification model to identify positive, negative, or neutral sentiment toward **Apple** and **Google**.
- Generate actionable, data-driven insights to support key business and marketing strategies

Key Outcomes:

- Successfully fine-tuned **Random Forest** and **XGBoost** models.
- Achieved a test accuracy of **98.21%** and a Macro ROC-AUC of **0.768** with the final **XGBoost** model.
- Demonstrated the feasibility of a scalable, real-time sentiment monitoring system.

Data Overview

SOURCE	CROWD FLOWER
9093	TWEETS
36%	PRODUCT MENTIONS
22	DUPLICATED VALUES

Key Data Constraint:

- The dataset is recorded as at 2013, which is **12 years in the past** from the time of this analysis.

Data Preprocessing

01

Lowercasing

All tweets were handled in lower case for uniformity.

02

Stopwords Removal

Stopwords and punctuation marks removal to leave only relevant text.

03

Tokenization and Lemmatization

to break words into units that are easier to analyze in their root form.

04

Imputation and Vectorization

To identify the most important and distinctive words in the corpus.

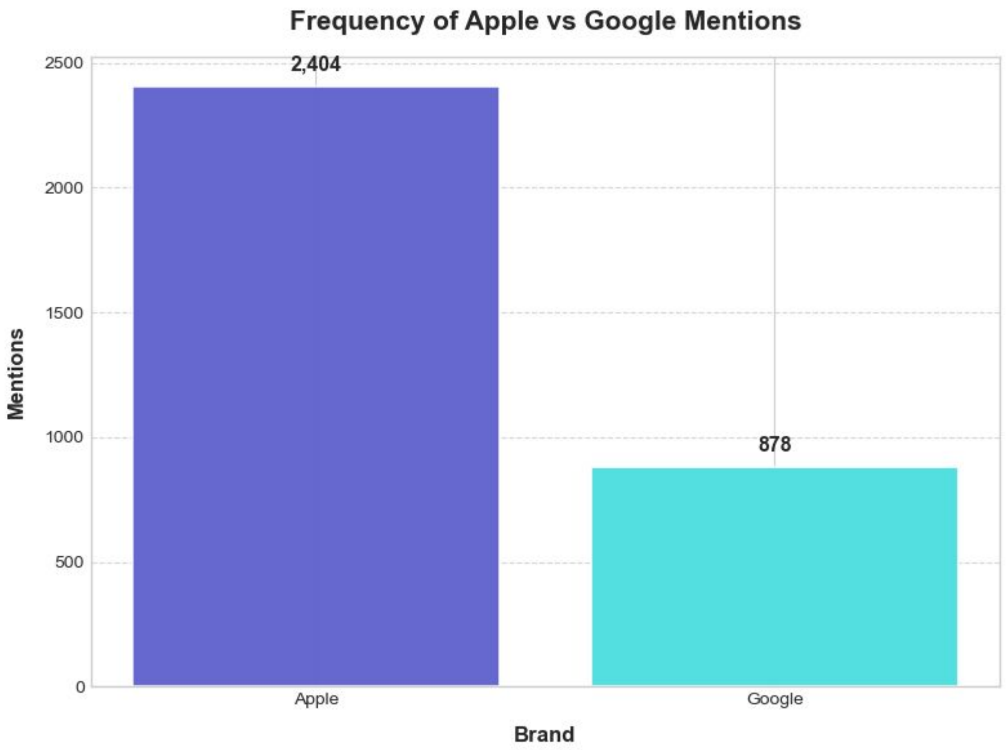
s.

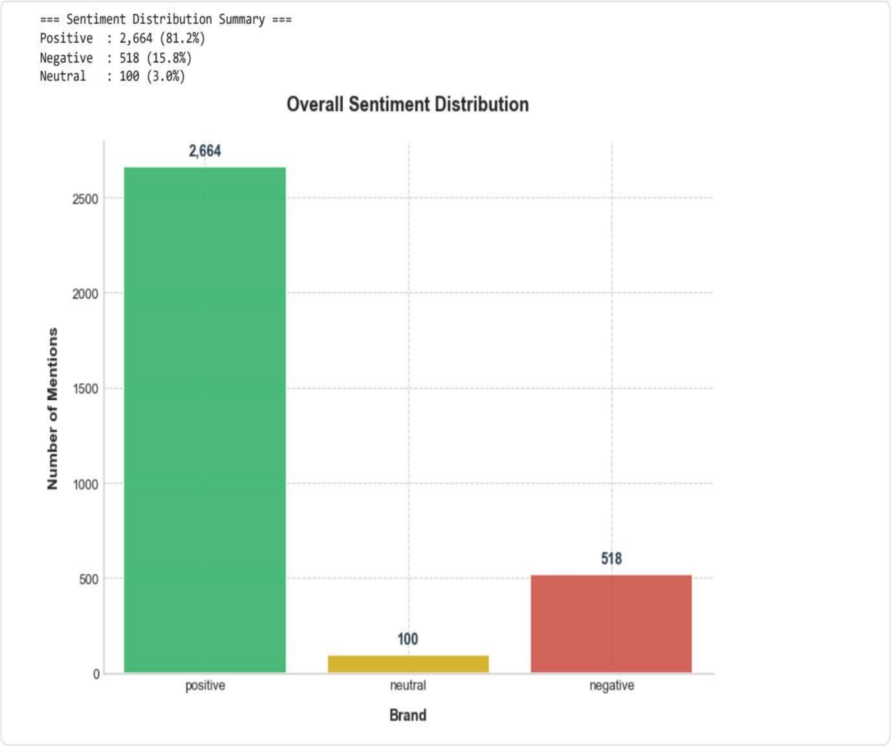
Overall Conversation Volume and Sentiment Distribution

Key Finding 1:

Frequency of Mentions

- **Apple is the most frequently mentioned entity**, signifying a higher volume of user commentary





Key Finding 2: Overall Sentiment

- Sentiment is **predominantly positive (81.2%)**.
- Negative sentiments are only **15.8%** of the data.

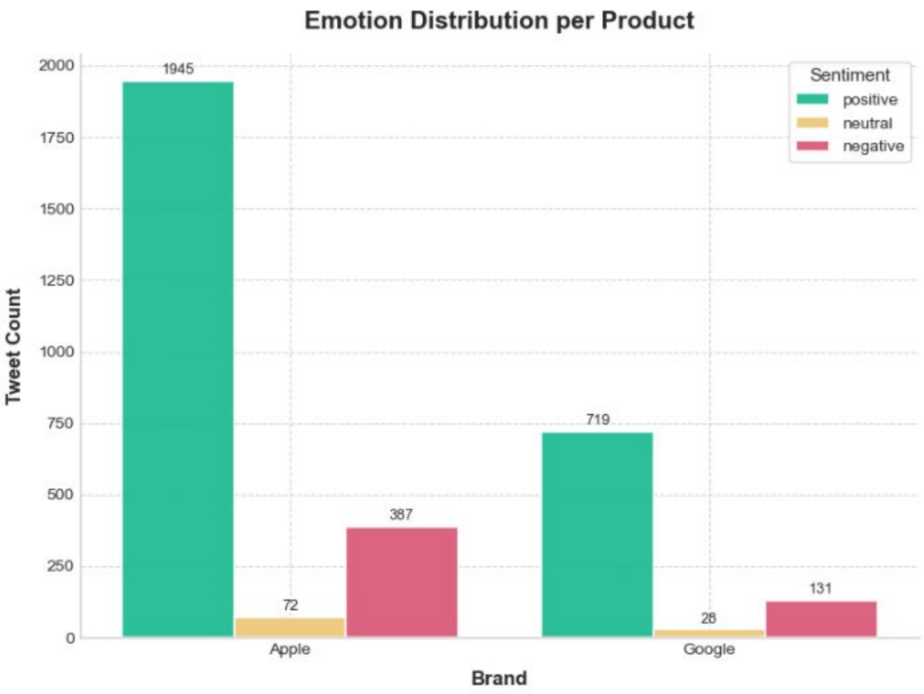
Apple vs. Google: Emotional Intensity and Product Focus

Apple's Focus: Product-Driven and High Emotional Intensity

- Dominates the discussion in both **volume and emotional intensity**.
- Conversations remain **product-driven** (*iPad, iPhone, store*) across all sentiment categories.

Google's Focus: Ecosystem-Driven and Lower Volatility

- Sees a **smaller and less volatile conversation**.
- Discussions center around its **broader ecosystem** (*Android, Maps, Social, Circle, Network*)



Thematic Differences: Retail Experience vs. Digital Innovation

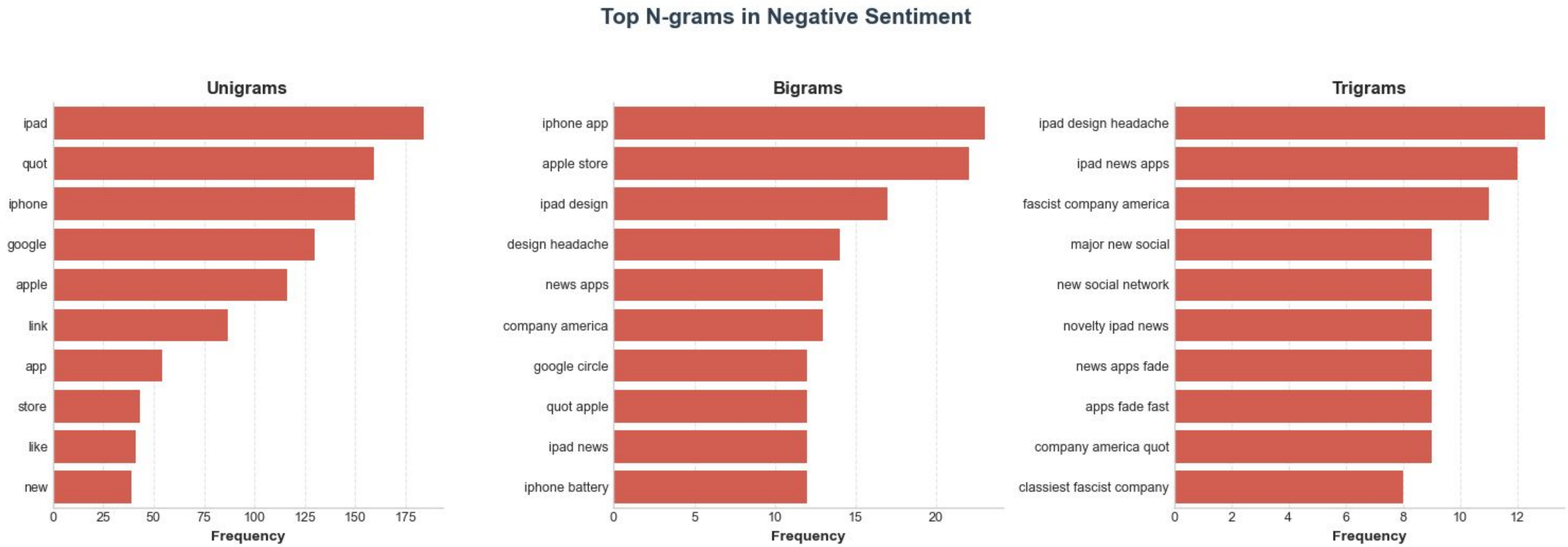
Apple's Narrative: Experiential & Retail

- **Top Trigrams:** *apple pop store*, *store downtown austin*.
- Anchored in **tangible experiences** and **design-centered discussions**.

Google's Narrative: Conceptual & Innovation

- **Top Mentions:** *social network*, *circle*, *launch major new*.
- Focuses on **future potential** and is more **conceptual/innovation-oriented**.

Top N-Grams in Negative Sentiment



Negative Sentiment Themes:

- **Apple:** *design headache*, *battery* (product frustration).
- **General:** *fascist company america*, *novelty apps fade fast* (brand fatigue/criticism).

Machine Learning Performance and Recommendation

Modeling Approach Summary:

- **Target:** Sentiment (Positive, Negative, Neutral).
- **Algorithms:** Random Forest and XGBoost (Ensemble focus).
- **Class Imbalance Handled By:** Class weighting and SMOTE.

Tuned Model Results (Key Metrics):

Metric	XGBoost (Tuned)	Random Forest (Tuned)
Testing Accuracy	**84.47%**	78.08%
Macro F1-Score	0.46	0.46
Macro ROC–AUC	**0.768**	0.705

Recommendation: XGBoost is the recommended model for deployment due to its **higher accuracy** and **superior ROC-AUC**.

Limitations and Path Forward

Class Imbalance Sensitivity

01

Minority classes (especially Neutral) have lower recall and precision despite class weighting.

Overfitting Risk

02

The gap between training (98.2%) and test (84.5%) accuracy indicates mild overfitting.

Interpretability Challenge

03

XGBoost decision boundaries are complex and opaque to non-technical stakeholders.

Future Improvement Areas

Better Imbalance Handling

Test advanced resampling strategies (e.g., SMOTEENN or ADASYN).

Model Monitoring

Continuously monitor model drift and retrain periodically using fresh data to maintain accuracy over time

Advanced Explainability

Implement SHAP or LIME for improved transparency.

Actionable Business Takeaways

Apple (Focus on Experience and Product Quality):

- **Action:** Use real-time sentiment monitoring for product launches and retail events to immediately address issues.
- **Insight:** Even negative sentiment revolves around flagship products (*iPhone*, *iPad*), signifying strong engagement but **high consumer expectations** for quality and design.

Google (Focus on Narrative and Emotional Appeal):

- **Action:** Improve **clarity and emotional appeal** in messaging around new platforms (*Circle*, *Social Network*) to foster more enthusiasm.
- **Insight:** Discussions are centered on the brand's **ecosystem and innovation** (*Android*, *Map*), framing it as a platform rather than just a product.

Thank you